APEX INTERNATIONAL ACADEMY

CBSE Senior Secondary Affiliation No. 1930482 | Academic Excellence & Research Division

OFFICIAL ACADEMIC SOLUTION & EVALUATION DOSSIER

VERIFIED & GRADED

ASSIGNMENT: **Microorganisms: Friend and Foe**

SUBJECT: General Science (SCI101)

CLASS & SECTION: Class 8 - Section A

ACADEMIC YEAR: 2025 - 2026

MARKS EVALUATED: **95 / 100 (Grade A)**

TEACHER DIRECTIVE & PROBLEM FOCUS:
Research bacterial fermentation in curd, yeast in baking, nitrogen fixation, and pathogen pathology.

1. Executive Overview & Problem Context

Microorganisms encompass microscopic single-celled or colonial living entities ubiquitous across Earth's biosphere. Categorized into Bacteria, Fungi, Protozoa, Algae, and Viruses, these agents maintain essential ecological nutrient cycling and industrial bio-processes, while concurrently posing pathogenic risks to human, animal, and plant vitality.

2. Core Analysis, Solutions & Systematic Working

- **Industrial Fermentation: Lactobacillus & Yeast:**

  Lactobacillus bacteria ferment milk lactose sugar into lactic acid, dropping pH and coagulating casein proteins into curd. Yeast (Saccharomyces cerevisiae) executes anaerobic respiration in dough: C †H • ,O † -> 2C ,H …OH + 2CO ,. Trapped carbon dioxide gas bubbles expand during baking, creating light, porous bread.

- **Nitrogen Cycle & Leguminous Bio-Fixation:**

  Atmospheric nitrogen (N ,, 78%) is biologically unavailable to plants. Symbiotic Rhizobium bacteria dwelling within root nodules of legumes (peas, lentils, clover) fix N , into soluble ammonium and nitrates, replenishing soil fertility naturally without synthetic chemical fertilizers.

- **Pharmaceutical Discoveries: Antibiotics & Vaccines:**

  Sir Alexander Fleming's 1928 discovery of Penicillin from Penicillium notatum mold revolutionized medicine. Antibiotics inhibit bacterial peptidoglycan cell-wall synthesis or protein translation. Vaccines introduce weakened or inactivated antigens to stimulate memory B-lymphocytes for long-term immune protection.

- **Pathogenic Microbes & Food Preservation Science:**

  Microbial foes cause infectious diseases: Tuberculosis (Mycobacterium tuberculosis), Cholera (Vibrio cholerae), Malaria (Plasmodium vivax via female Anopheles mosquito vector). Food preservation techniques (Pasteurization at 72°C for 15s, vacuum canning, dehydration, salting, chemical sodium benzoate) eliminate spoilage.

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3. Detailed Data Matrix & Systematic Findings

Microbe Name	Domain / Class	Beneficial / Pathogenic Role	Key Scientific Impact
Lactobacillus acidophilus	Bacteria	Beneficial (Fermentation)	Curd production & gut probiotic microbiota
Saccharomyces cerevisiae	Fungi (Yeast)	Beneficial (CO , Leavening)	Baking bread & brewing fermentation
Rhizobium leguminosarum	Bacteria	Beneficial (Symbiosis)	Biological nitrogen fixation in root nodules
Plasmodium vivax	Protozoa	Pathogenic (Malaria)	Ruptures human erythrocytes (red blood cells)
Mycobacterium tuberculosis	Bacteria	Pathogenic (Tuberculosis)	Infects pulmonary alveoli through aerosol droplets

4. Student Conclusions & Reflective Synthesis

Microbiology illustrates the duality of nature: harnessable microbial metabolism underpins biotechnology and agriculture, while rigorous hygiene, sterilization, and vaccination guard public health against virulent infectious vectors.

5. Official Faculty Evaluation & Grading Rubric

ASSESSMENT CRITERIA BREAKDOWN

Conceptual Clarity & Factual Accuracy: 28 / 30

Problem Solving & Structured Working: 18 / 20

Analytical Depth & Research Methodology: 28 / 30

Presentation, Organization & Neatness: 14 / 20

FACULTY FEEDBACK REMARK:

"Outstanding analytical depth & presentation! Microbial taxonomy, biochemical pathways, and epidemiological vectors are thoroughly explored."

APEX EVALUATION DIVISION

VERIFIED & ARCHIVED